

YG IAspire 2.0

Title : *“Prarochana AI: AI-Enabled Ad Campaign Generation and Analysis “*

Theme: Others, AI related

Team Members: *Ganta Srilalitha (Empld: 31590)*
Rithik Gundeti

Problem/Opportunity

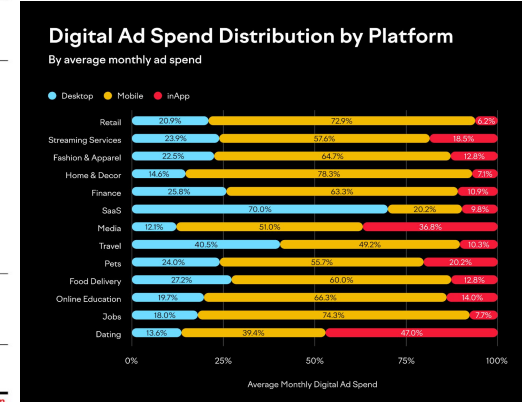
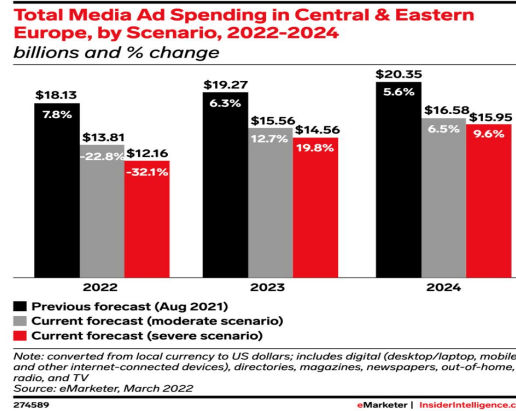
a. What problem are we solving?

Ad Campaign Generation Challenges

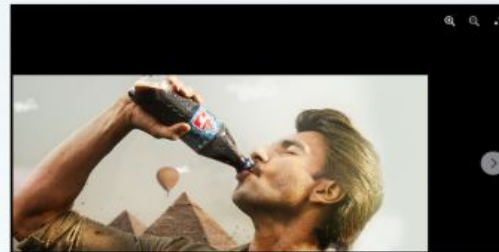
- Costs Money
- Too Many Campaigns!!
- Inability to Real-Time Fine-Tune
- Feasibility of Campaign Execution

b. Who is impacted?

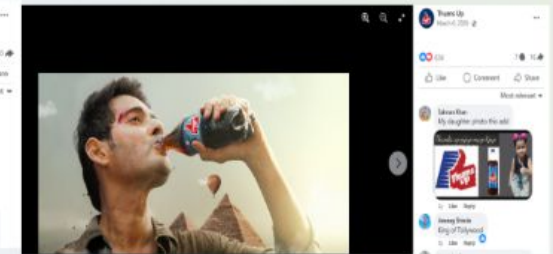
- Marketing & Brand Management
- Digital Marketing Teams
- Advertising Agencies
- FMCG Brands



“Real-time fine-tuning wrt to competitor companies and market trends”.



305 fb reactions



604 fb reactions

Proposed Solution

a. Idea Overview

Prarochana AI is an AI-powered marketing platform that generates **complete advertisement campaigns by combining Generative AI, Retrieval-Augmented Generation (RAG)**, and campaign analytics.

The solution generates:

- **Ad captions**
- **Advertisement images**
- **Budget recommendations**
- **Campaign insights**

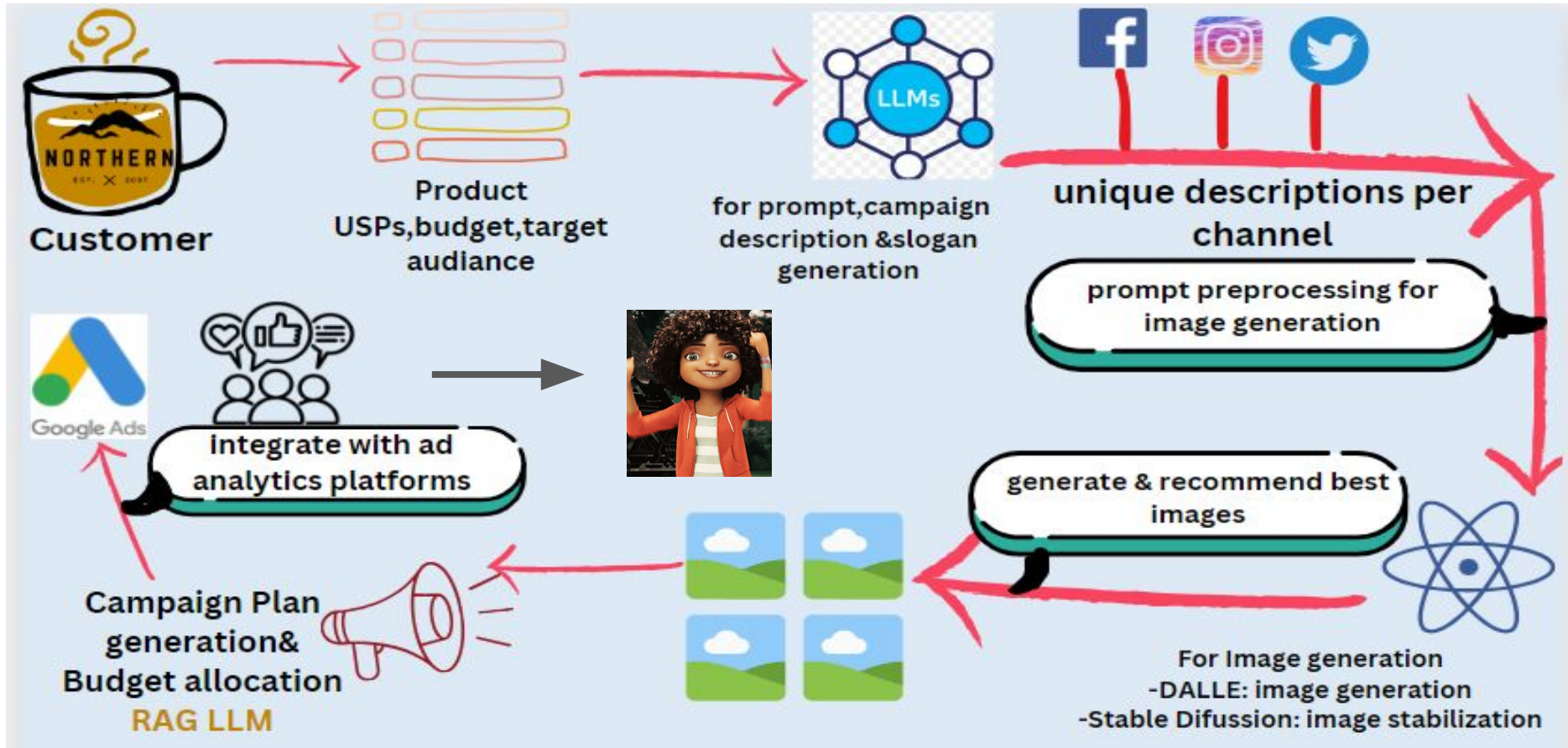
based on product details and target audience inputs.

Company	Beverage Type	Campaign name	Slogan	Year	Msg Conveyed	Product Description	Product Price	Competitor companies	Ambassador	Ambassador traits	Target Audience
Frooti	Mango juice	FrootiLife	ie Real Fruit Jul	2022	To showcase the natural and authentic taste of Frooti	A mango juice drink made from fresh mango pulp and no added sugar or preservatives	Rs. 15 for 200 ml	Maaza, Slice, Paper Boat	Alia Bhatt	Young, trendy, bubbly	Health-conscious and quality-seeking consumers

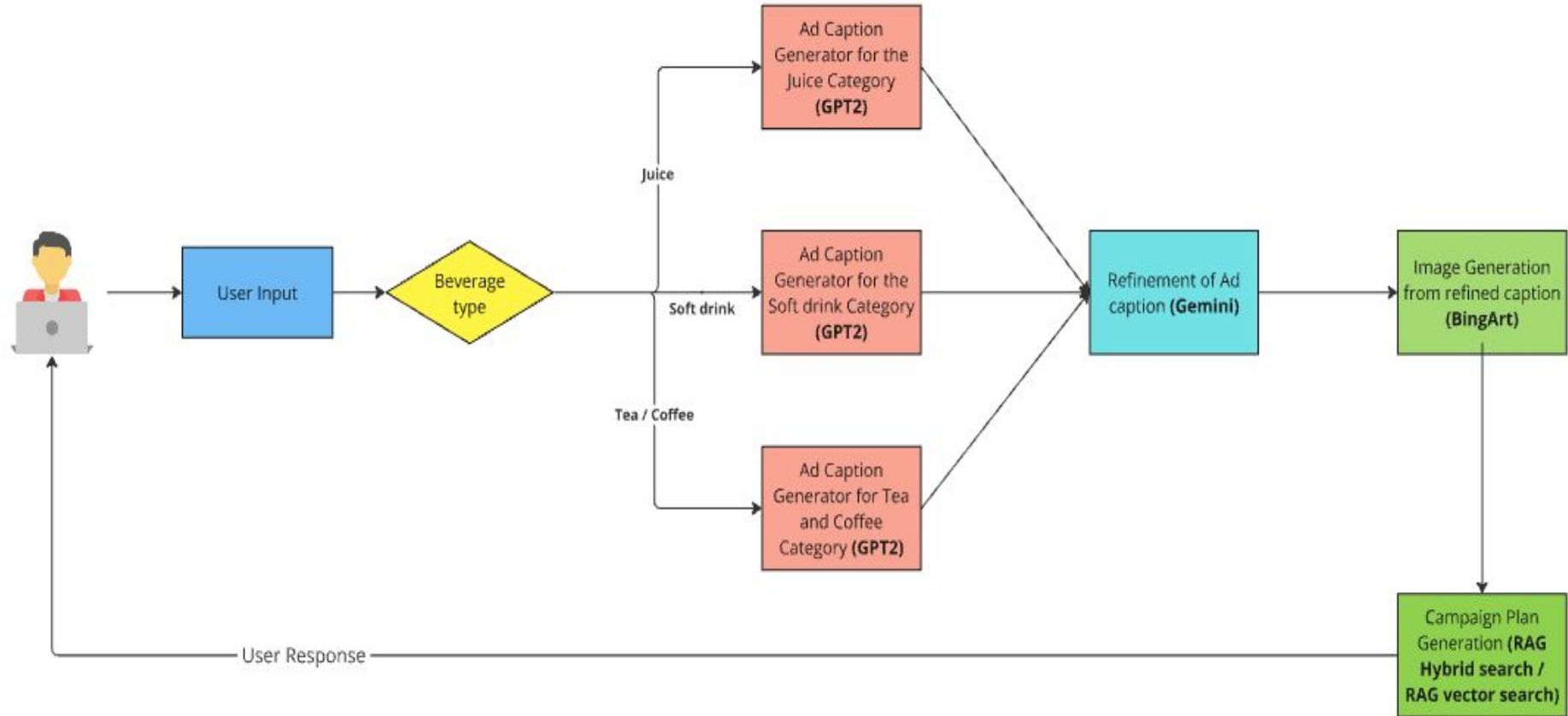
b. Key Features / Components

- **Dataset Integration** : Curated and integrated the AdBits dataset comprising rich metadata from Indian beverage ad campaigns, including campaign **goals, audiences, prices, slogans, ambassadors**, and media spend.
- **Text Generation Models** : Fine-tuned transformer-based models (**GPT2, Gemini**) using SFT with LoRA to generate creative ad captions from structured inputs like product description, pricing, and target demographics.
- **Text-to-Image Generation** : Used **DALL-E** -based models to generate visual ad mockups from the generated ad caption prompts, translating textual ideas into ad visuals.
- **Retrieval-Augmented Generation (RAG) for Budgeting** : Implemented a RAG pipeline that retrieves budgeting patterns from similar past campaigns to suggest optimal media-wise budget allocations.
- **User Interface** : Developed a **Streamlit-based** interface where users input campaign parameters and receive AI-generated ad captions, visuals, and budget breakdowns in real-time.

Architecture



End to end workflow for Beverage Ad campaign generation



Demo:

- https://drive.google.com/file/d/1QNjOvSULOUFzXqES8zezEETSp-tteUM7/view?usp=drive_link

IEEE Publication:

- <https://ieeexplore.ieee.org/document/11077247>

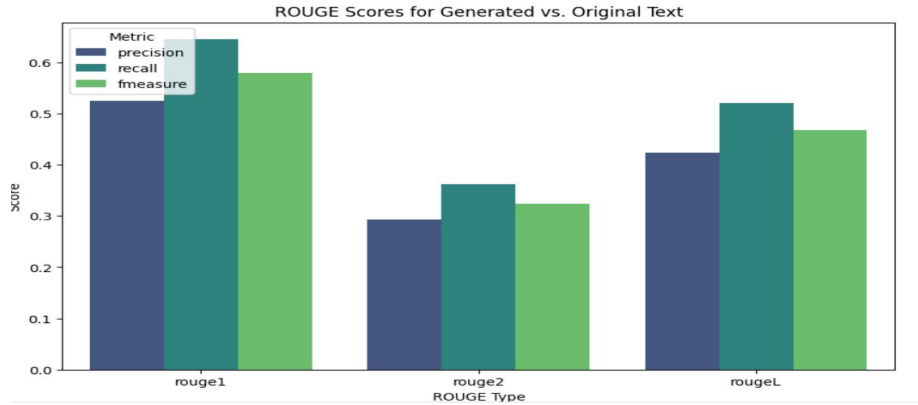


Best Paper Recognition.jpg

Rewarded with Best Research Paper (Prarochana AI: AI Enabled AdCampaign Generation and Analysis) under AI-ML Track at ICCC-6th Edition IEEE Conference held at College of Engineering, Trivandrum, Kerala

Results Obtained

ROUGE SCORES OBTAINED FOR TEXT GENERATION USING LLMs:



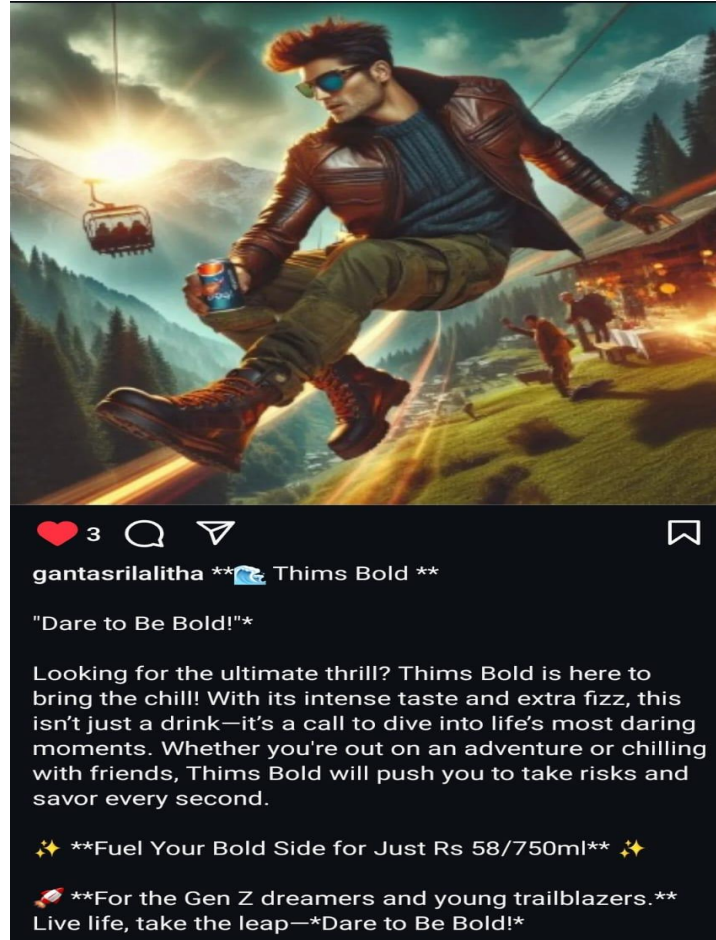
The ROUGE-1 score obtained was **0.57** , and
The ROUGE-L score obtained was **0.46**.

CLIP SCORES OBTAINED FOR AD IMAGES GENERATED:

- The best clip score we were able to generate is **40.301** , which is for an ad image generated by the Microsoft BingArt
- **Stable Diffusion** and **DALL-E** gave clip-scores of **38.45** and **33.609** respectively.
- CLIP Score is an embedding based metric that can be used to evaluate the correlation between a generated caption for an image and the actual content of the image.

Results Obtained

We pushed our Advertisement AI images to social media platforms like Instagram :



Expected Benefits

Business Benefits and Qualitative / Quantitative Impact

- ▶ **Efficiency in Ad Creation:** Prarochana AI streamlines the advertisement campaign creation process using various GenAI models, significantly reducing the time, cost, and manual effort traditionally required.
- ▶ **Targeted Approach:** The system ensures ad captions and images align closely with the target audience and product specifications, enhancing engagement and effectiveness.
- ▶ **'Prarochana AI' application:** Streamlit integration enabled a user-friendly application, effectively integrating backend AI models to frontend functionalities.
- ▶ **Testing:** Qualitative testing showed a preference for AI-generated ads among users, with 74% of the surveyed population finding them more appealing.

Feasibility & Risk

a. Effort

7-8 MonthsDevelopment includes:

- Dataset preparation
- LLM fine-tuning
- RAG implementation
- Image generation integration
- Frontend development
- Testing and validation

The solution is modular, making it scalable to other FMCG categories.

b. Dependencies

- Large Language Models (GPT2, Gemini)
- Text-to-image models (DALL·E/BingArt)
- Pinecone Vector Database
- MongoDB Atlas
- Campaign dataset (AdBits)
- Streamlit application framework

c. Risks

Risk	Mitigation
AI-generated content quality	Fine-tuned models with domain-specific dataset
Budget prediction accuracy	RAG with historical campaign retrieval
Dataset limitations	Continuous dataset expansion

Recommendation & Next Steps

a. Pilot Scope

Launch a pilot with:

- Beverage companies
- FMCG marketing teams
- Advertising agencies

Use cases:

- Campaign generation
- Budget recommendation
- AI creative generation
- Campaign performance analysis

b. Immediate Actions

- Expand AdBits dataset across more industries
- Integrate real-time market trend analysis
- Validate with enterprise marketing teams

c. Decision Required

Seeking approval to:

- Expand the solution beyond beverages into broader FMCG categories
- Invest in additional datasets and AI infrastructure
- Validate business ROI through industry partnerships

